

MIDTERM EXAM

ECON 2010 – Macroeconomics

Spring 2019

To be completed in-class on Wednesday, March 13, 2019. You will have 2.5 hours to complete the exam (5:30-8:00pm).

Name:

Midterm exam instructions:

1. There are **three** questions on the exam, each with multiple parts. Please complete all parts of all questions.
 2. Read each problem carefully. Just because it looks like a problem you've seen before doesn't mean that the exam asks you to do exactly the same thing in the problem that you've done before.
 3. Pay particular attention to when a question asks you to **explain** your answer in words, and when you do not need to. Explanations are usually worth more than other types of problems because they show deeper understanding – i.e. not just *what* to do, but *why*.
 4. Write your final answers as legibly as possible in pen. Make sure that it is clear to me what your final answer is.
 5. Show your work. Even if your answer is not exactly correct, you will receive credit for setting up the problem correctly. The better that I understand what you've done and why, the more likely you are to receive partial credit.
 6. You may use a calculator, but still show work in terms of how you set up the problem step by step.
 7. You may not use notes, the textbook, any other materials on the exam, or confer with anyone about the exam while you're taking it. To say it differently, this is a “closed book” exam.
 8. Please do not ask the exam facilitator questions about what you're supposed to do to answer a given question, or what I intended when I wrote a question. Instead, if you think that a question is ambiguous, make a choice in how you interpret the question so that you can answer the question, and *explain your interpretive choice in your answer*. I will take into account your interpretation (if I agree that there is ambiguity) when grading.
 9. Take a deep breath, remember how much you've learned so far, and do your best!
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DO NOT TURN THIS PAGE UNTIL INSTRUCTED TO DO SO.

Problem 1 — Thinking Like an Economist: Considering the US-Mexico Border Wall (20 points)

In this problem you will consider how economists have employed various principles of economic thinking in analyzing the proposed border wall between the US and Mexico.

For each statement below, circle **one** of the two principles of economic thinking that is being utilized/applied in the statement. Although you may think that both principles are being applied, please do your best to choose the principle that *best* represents the *main* principle being utilized. I am not trying to trick you with these statements, although you may find them challenging.

(a) *“The simple problem is drugs are tremendously profitable. That makes it so, no matter how many barriers (physical or otherwise) the US puts up, someone will always try to find a way around the barriers and ship the drugs anyway. There’s just too much money to make in this space for criminal organizations to not try to take part.” (4 points)*

- (A) Rational actors (B) Equilibrium

(b) *“Consider just the number of illegal crossings that have taken place just along the western Texas-Mexico border in the past two years. This is just one small stretch of border, and just one relatively short period of time, but it suggests the general conditions that border crossers face in terms of environmental hardship and risk of apprehension by the authorities.” (4 points)*

- (A) Rational actors (B) Simplifying assumption

(c) *“One of the debates being had is what the money that President Trump has claimed through executive authority by declaring a state of emergency would be spent on if it were not spent on the wall. The President has claimed that money he is appropriating from the defense budget is in effect, defense department slush funds — not earmarked for any particular purpose, and therefore unlikely to take resources away from important projects. Others disagree, noting that the defense budget may not specify a particular use for funds, but that funds are allocated to such highly specific departments, units, or geographies that taking these funds away to build a border wall could substantially disrupt other military operations. Because of this, the true consequences of shifting this money from one purpose to another are quite significant.” (4 points)*

- (A) Normative thinking (B) Opportunity cost

(d) *“Were you to let free migration across borders, what would eventually happen, the economist explained, is an equalization of wages between countries. Migration occurs, she said, because wages are higher in the US than in Mexico. Were free migration allowed, you would have people migrate until you reached a situation in which there was no reason for anyone in the US to move to Mexico and no reason for anyone in Mexico to move to the US because they couldn’t make themselves better off by moving. If the wage is the same here as it is there, no one has a reason to change what they’re doing and pack up and move, she explained.” (4 points)*

(A) *Ceteris parabis*

(B) Equilibrium

(e) *“Were the border wall to reduce migration (a question in its own right), the economists estimated that the effect of reducing illegal migration from Mexico by 1% would result in just a few cent increase in annual wages for low-skilled workers in the US, no meaningful change in wages for high skilled workers, and a small reduction in US GDP.” (4 points)*

(A) Opportunity cost

(B) Thinking on the margin

Problem 2 — The Puzzle of Unemployment (49 points)

In this problem you'll discover that there is a puzzle to why unemployment exists *at all* in an economy.

To make this discovery, we will consider a simple economy called Greenland, where there is only one industry — ice sculpture making. We'll specifically consider the ***labor market*** for ice sculptors in this problem.

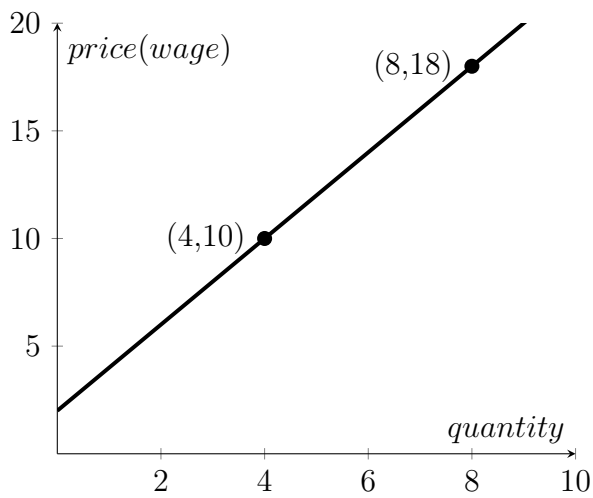
There are **10 ice sculptors** (i.e. individual laborers) in Greenland, and **2 firms** that the 10 sculptors can potentially work for. In Figure 1, I've drawn for you the labor market supply curve for ice sculptors, the labor market demand curve for ice sculptors, and a plot that puts both together to find the equilibrium in this labor market.

IMPORTANT. Take note of a few measurements and assumptions that you will need to analyze this economy.

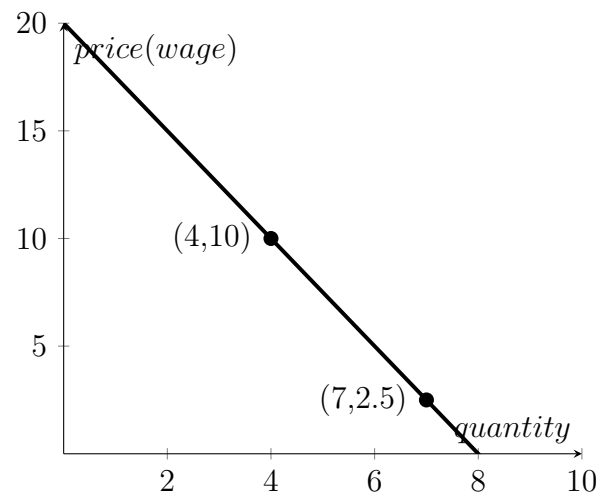
- The price (wage) in this labor market is measured as an *hourly wage*.
- The quantity is measured in *hundreds of total hours worked per week* (that means that a 1 on the x axis indicates one *hundred* total hours of work; a 2 on the x axis indicates two *hundred* total hours of work, and a 10 on the x axis indicates 1,000 total hours of work, etc.).
- Assume that each laborer can supply at minimum 0 hours of work, and at most 100 hours of labor per week.
- Assume that there are no laws or contracts that *require* a minimum number of hours of employment for ice sculptors in Greenland, nor any laws or contracts that *require* all 10 of Greenland's laborers to work an *equal* number of hours.
- Assume that all workers are 16+ years old.
- Assume that anyone who loses their job in Greenland looks for work (i.e. remains in the labor force).
- Hold all else fixed when you analyze any change you are directed to consider below.

Figure 1: Market supply, market demand and equilibrium for ice sculptors

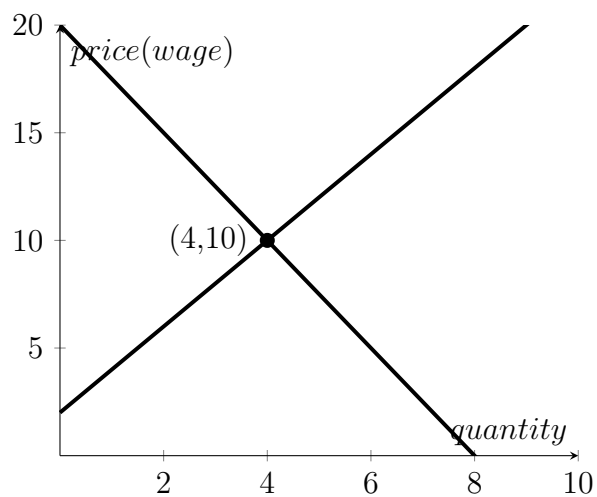
Plot A: Market supply curve



Plot B: Market demand curve



Plot C: Market supply and market demand, a.k.a. market equilibrium



- (a) Consider Plot A in Figure 1 above. The plot indicates that were the wage to decrease from \$18 to \$10 per hour, the total quantity of labor supplied in the market would decrease from 800 to 400 hours per week.

True or False: One possible reason for this *specific* change in the number of hours of labor supplied is that a new industry has started on the island, seal training (they are trained to balance beach balls on their noses), which offers high wages and excellent benefits. In just a few sentences at most, **explain briefly why the statement is true or false. (5 points total; 1 point for T/F, 4 points for the explanation)**

- (b) Consider Plot B in Figure 1 above.

True or false: It *must* be the case, given the market demand curve, that at the wage \$2.50, each of the two firms in the ice sculptures market demands exactly 350 (i.e. $\frac{700}{2}$) total hours of ice sculptures making labor. In just a few sentences at most, **explain briefly why the statement is true or false. (5 points total; 1 point for T/F, 4 points for the explanation)**

- (c) What is the equilibrium in the ice sculpting labor market indicated in Plot C of Figure 1? **(5 points)**

- (d) If all 10 workers in Greenland are employed in the ice sculpting labor market, give **two different** examples of how many hours the 10 workers could **each** be working, **in equilibrium**, according to Plot C. To do so, simply fill in the table below with your two examples. **(4 points)**

Worker	Example 1 Hours	Example 2 Hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

- (e) Consider the equilibrium in the ice sculptures making labor market that you described in part (c) above. Analyze what happens to the market when a **new ice sculptures making robot** is introduced to the market. Assume that the robots make ice sculptures faster and better than human sculptors. Therefore, it is cost effective for both of the two sculpture firms to purchase *at least a few* of these robots. To be very clear, the robot does not help workers make ice sculptures better or faster, it is a substitute for their labor. Remember to hold other factors fixed (*ceteris parabis*) as you perform your analysis. **(9 points total)**

(1) Which curve(s) shift as a result of this change? You could say either labor supply, labor demand, none, or both. **(2 points)**

(2) For each curve that you indicated would shift in part (f1) above, indicate which direction the curve will shift (**either right or left**). (If no curves will shift, just say “no curves will shift.”) **(2 points)**

(3) **Draw** a new market equilibrium plot that contains the curves in Plot C and **also** reflects the curve shifts that you think would occur according to parts (f1) and (f2) above. **Draw** the previous equilibrium curves in solid lines, and the shifted curve(s) in dashed lines. (If no curves shift, just draw the starting solid curves from Plot C and write “no curves shift.”) **State** how the wage and quantity of labor demanded have changed relative to the equilibrium indicated in Plot C — are each higher, lower, the same, or unknown as compared to their initial equilibrium levels?

(Note that I do **not** give you enough information to put specific numbers on the new equilibrium or the changes. Also, you do **not** need to draw the starting equilibrium as precisely as in Plot C (this is not a drawing test!); only well enough for you to show the shifts in the curves you may make, and analyze the equilibrium change.) **(5 points)**

- (f) Think very carefully: Can the equilibrium change that you found in part (f) above have occurred **without** a change in the *unemployment rate*? Put differently: Can the unemployment rate have stayed the same after the equilibrium change you considered in (f) above? **Yes or no?** In just a few sentences at most, **explain briefly why or why not**. Hold other factors besides the shift in curves you found fixed for this analysis (*ceteris parabis*). (If you did not find an equilibrium change in part (f), just state “no change.”) **(5 points)**

- (g) Consider the plot that you drew in part (f3) above, but make a new assumption. Assume that the wage rate is “stuck” or “fixed” at the equilibrium level given in Plot C above. That is, the wage *cannot*, I repeat, *cannot* fall below \$10 per hour. The wage can rise above \$10 per hour, but it cannot fall below. However, the quantity of labor supplied and the quantity of labor demanded *can* potentially change. That is, the wage is “stuck”/“fixed”, but the quantity is not.

Re-draw the full plot that you drew with both original and any shifted curve(s) from (f3) here. **Label** the starting equilibrium (at the point where the initial curves cross) with an **A**. Next, **label** the point that results from both any shifted curve(s) *and* from the stuck wage with a **B**. On the quantity axis, mark the quantity of labor demanded in the starting equilibrium and **label** it Q_A . On the quantity axis, mark the new quantity of labor demanded that results from any shifted curve(s) and the stuck wage and **label** it Q_B . On the price/wage axis of your plot, mark the wage for equilibrium A and **label** it W^A . On the price/wage axis of your plot, mark the wage for point B and **label** it W^B . If you think some of your points have two labels, that’s fine. If you did not draw shifted curves, just re-draw your (f3) plot from above and write “no change”. **(6 points)**

- (h) The difference between **which two points** that you labeled in your plot for part (h) above indicates a market with excess labor supply? **(4 points)**

- (i) Let's relax one of the assumptions from the start of the problem and consider what would happen if every one of Greenland's ice sculptors has an individual contract that entitles them to 40 hours per week of work at \$10 or more per hour ***if they are employed***. Again, let's assume that *if* they are employed, they have a contract with the above guaranteed wage and hours-per-week, but they don't have to be employed. How does the unemployment rate change as a result of the movement from equilibrium A to B in part (h) above? Does the unemployment rate go **up, down, or stay the same**? Assume that the firms honor these contracts, and that there are no other changes in the economy that happen at the same time (hold all else fixed). In just a few sentences, **explain briefly your answer**. (6 points)

Problem 3 — Making the Most out of Living with a Roommate (26 points)

Imagine a situation in which you're living with a roommate. There are a bunch of household chores that you each have to do each week to be successful adults. Principle among these chores are (1) doing the laundry, and (2) doing the dishes. Unfortunately, you don't have a washer, drier or dish washer, so you have to perform these chores all by hand. What a pain.

You and your roommate each have specific speeds at which you can do each of these tasks. Table 1 below indicates how fast each of you can perform these household chores in a given hour.

Table 1: Productivity in performing household chores (per hour)

	Dishes washed	Clothes washed
You	20	5
Roommate	60	10

(a) Who has the absolute advantage (you or your roommate) in doing the dishes? **(1 point)**

(b) Who has the absolute advantage (you or your roommate) in doing the laundry? **(1 point)**

- (c) What is your opportunity cost of washing dishes in terms of washing clothes? Write your sentence to answer this question in the following way: “I have to give up washing _____ dishes in order to wash _____ clothes.” **(2 points)**
- (d) What is your roommate's opportunity cost of washing dishes in terms of washing clothes? Write your sentence to answer this question in the following way: “My roommate has to give up washing _____ dishes in order to wash _____ clothes.” **(2 points)**
- (e) Who has the *higher* opportunity cost of washing dishes? Put differently: Who has to give up washing more dishes to wash clothing? **(3 points)**
- (f) Who has the *higher* opportunity cost of washing clothes? Put differently: Who has to give up washing more clothes to wash dishes? **(3 points)**

- (g) Who has the comparative advantage in washing dishes, and therefore, should specialize in washing dishes? You or your roommate? **(3 point)**
- (h) Who has the comparative advantage in doing laundry, and therefore, should specialize in doing laundry? You or your roommate? **(3 point)**
- (i) Before you and your roommate got to know each other, you each did your *own* laundry and you each washed your *own* dishes. You each spent two hours each week washing things; one hour washing clothes, and one hour washing dishes. Given this, how many total items (the *combination* of both clothes and dishes) were washed in your “household economy” in a given week when you each did your own laundry and dish washing? (Provide me **one** number of total items washed.) **(4 point)**

- (j) After you and your roommate became friends, you decided to employ the principle of comparative advantage and specialize in tasks. Given how you identified that you each would specialize in previous parts of the problem, how many total items can you wash if you *each* still spend 2 hours each week washing, but you now do only the washing that you have the comparative advantage in? (Provide me **one** number of total items washed.) **(4 point)**
- (k) What is the difference in total items washed per week between the first situation where you and your roommate do *not* specialize, and the second situation where you and your roommate *do* specialize? Do you produce more or less *in total* with specialization? **(1 point)**